The phosphosites of one protein

Every phosphosite measured on one protein, against every sample.

AT A GLANCE

Where	Single protein → Heatmap
Data levels	Phosphosite only: the dataset-level selector is fixed on this tab and the page says so.
Needs	A phospho dataset carrying PTM_Collapse_key and a protein-group column, and one protein selected.
Data source	Data, Filtered Data or Imputed Data; the page opens on the most processed one that exists.
Rows, columns	One row per phosphosite of the selected protein; one column per sample of the selected conditions.
Two scales	**Value Type** log2 colours the stored values; z colours each row's deviation from its own mean.
Table	Yes, below the plot, downloadable as CSV — the matrix as drawn.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

One protein can carry twenty phosphosites and the boxplot draws one at a time. This page draws them all at once, against every sample, so the question it answers is which measured sites have similar patterns. It cannot separate a change in total protein abundance from a change in phosphorylation occupancy without additional measurements.

The cost is that a cell states only a colour. Nothing here is a test and nothing is aggregated by condition — a column is one run — so this page is for finding the pattern, and the boxplot or the volcano for the number that goes in the figure legend.

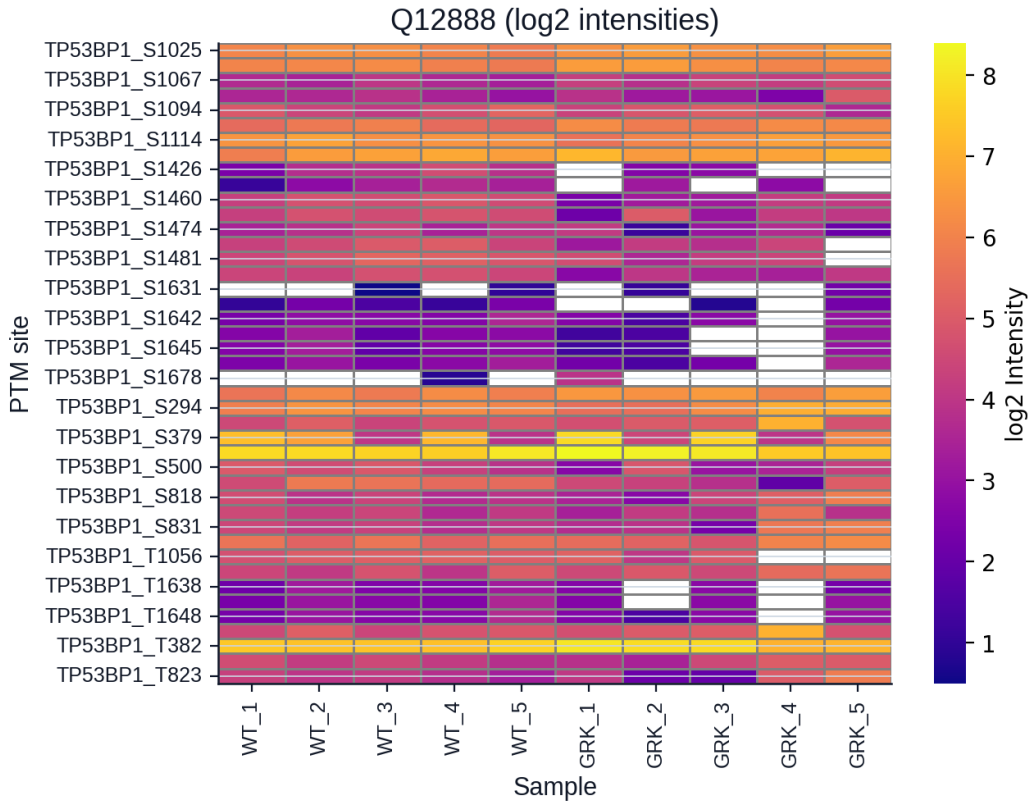


Figure 1.1. The phosphosites of protein group Q12888 — TP53BP1 — against the ten samples of the example dataset, five WT and five GRK, in **Value Type** log2 and with **Filter** *_**M1** at its default. That leaves 43 rows, which is why the row labels carry no multiplicity suffix. Colour is the stored value on the plasma scale at the right, and the white cells are the 38 values of this 43 × 10 matrix that are missing, several of them in the last two GRK runs. There are more rows than the figure has room to label, so every other site name is printed.

Reading it

- A cell: one phosphosite in one sample, coloured by that value on the scale **Value Type** chose. A cell left in the background colour is a missing value.
- A row: one phosphosite, labelled with the short site name — the accession is dropped because every row belongs to the same protein, and the multiplicity suffix too when **Filter** *_**M1** has already restricted the view to it. The whole identifier stays in the table.
- A column: one sample, labelled condition_n, numbered within its condition, with the sample's own identifier added on a second line when **Toggle ID** is on.
- The colourbar: log2 Intensity or Z-score, according to **Value Type**. It is the only thing that tells the two modes apart in an exported figure.

EASY TO GET WRONG

The two value types answer different questions. log2 colours the values as they stand, so an abundant site is bright in every sample and one a hundredfold below it is dark in every sample: the figure shows which sites are abundant, and a real regulation of the faint ones is invisible. z centres each row on its own mean, which makes the rows comparable and destroys the difference in level between them — a site whose values vary by a hundredth of a $\log_2$ unit gets the same full range of colour as one that moves fourfold. Read z for the pattern and log2 for the amount, never either for both.

2 Controls

Controls on this page

Control	What it does	Default	Where
Data source	Which matrix the values are read from: Data, Filtered Data or Imputed Data. Here the choice can change which *rows* are drawn, because clustering uses complete rows only.	the most processed matrix available	page
Identifier	Which name the protein is chosen by: Protein Names — the protein-group column of the phospho dataset — or Gene Names where the dataset carries a gene column.	Protein Names	page
Protein / Site	The one protein whose sites are drawn. The match is exact, so a stale selection draws nothing and says so rather than drawing a neighbouring protein group. Labelled **Gene** under Gene Names.	the first entry	page
Conditions	Which conditions contribute columns, and in which order.	all	page
Value Type	log2 colours the stored values; z colours each row's z-score across the selected samples.	log2	page
Color Map	The gradient the cells are painted with: plasma, viridis, magma, inferno or coolwarm. The appearance panel's own gradient overrides this one when it is set.	plasma	page
Toggle ID	Adds the sample's own identifier under each column label, on a second line.	off	page
Filter *_M1	Keeps only the rows whose PTM_Collapse_key ends in _M1, that is the singly modified form of each site, and shortens the row labels accordingly.	on	page

Controls on this page (continued)

Control	What it does	Default	Where
Cluster Rows	Orders the rows by hierarchical clustering instead of by the order they occur in the matrix.	off	page
Cluster Columns	Orders the columns by hierarchical clustering instead of by condition, so the grouping then has to be read off the labels.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20, 12, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Show title	The plot's own title is the selected protein with (log2 intensities) or (Z-transformed) after it; text, size, weight, colour and position are set here too.	shown	appearance
Colour scale	The gradient the matrix is painted with. Setting it overrides **Color Map** on the page; an unrecognised name falls back to the page's choice rather than failing.	unset, so **Color Map** applies	appearance
Axis labels, ticks, font, background	The Sample and PTM site labels, the row and column ticks, the font, and the paper colour — which is what a masked cell is drawn in.	plot's own	appearance
Condition and sample names	Offered, but this figure builds its column labels from the annotation and does not read the overrides. There is no legend card and no condition-colour card: the figure has a colourbar, not a legend.	the annotation's names	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

THE MATRIX

The rows of the phospho matrix whose protein-group column equals the selection are taken — by gene, those whose gene column contains the selected token. With **Filter** *_M1 on, only the rows whose PTM_Collapse_key ends in _M1 are kept. The columns are the samples of the selected conditions, relabelled condition_n in the order the annotation gives.

Every entry is coerced to a number; anything that is not one, and ±∞, becomes NA, and a row missing in every selected sample is dropped before anything is drawn.

SCALING

Value Type log2 colours the values as they are stored, which on this page is the $\log_2$ intensity, and the colourbar is captioned log2 Intensity.

Value Type z replaces every row by its z-score across the selected samples: for site i and sample j,

$$z_{ij} = \frac{x_{ij} - \bar{x}_i}{s_i},$$

where $\bar{x}_i$ and s_i are the mean and the standard deviation of that row over the samples in which it has a value, s_i with the $n - 1$ denominator. Both are computed *after* the conditions have been selected, so the centre of a row is the centre of what is on screen and adding a condition moves every colour in it. A row whose values are all equal has $s_i = 0$, cannot be scaled, and is removed rather than drawn flat.

CLUSTERING

With **Cluster Rows** or **Cluster Columns** on, the order is taken from hierarchical clustering with **average linkage** on **Euclidean** distances, computed on the matrix as scaled — distances between rows for **Cluster Rows**, between columns for **Cluster Columns** — and the plot is reordered to the leaves of the resulting tree. No dendrogram is drawn and no cluster boundaries are marked: the result is an ordering, not a partition, so nothing here states where one group of sites ends and the next begins.

A distance is undefined where a value is missing, so before either clustering runs *every row with a missing value in any drawn column is removed* — from the clustering, and from the figure and table with it. A notice beside the controls says so whenever clustering is on and the data source is not the imputed matrix. If no complete row survives, the error names the count and the two ways out: impute and select Imputed Data, or turn clustering off. Nothing switches the data source by itself.

Edge cases

- With one row, or one column, that clustering is skipped; the switch can be left on without effect.
- In z with a single sample selected a row's standard deviation is undefined, every row is removed, and the page reports an empty matrix. Use log2 to look at one sample.
- A missing value is masked: the cell is drawn in the background colour, with no colour from the scale — unless clustering is on, in which case its whole row has gone.
- **Filter *_M1** on a dataset whose collapse keys carry no multiplicity suffix removes every row, and the page reports that none remain after filtering. Turn the switch off for such datasets.
- On Imputed Data the cells that were missing carry imputed values instead and none are masked; the title does not say so, so label the figure yourself when it leaves the tool.

The filter stays on by default. On a dataset whose collapse keys carry no multiplicity suffix it therefore empties the heatmap and raises an error, and turning it off is the fix.

4 The table below the plot

The matrix exactly as drawn: the rows in the figure's order, scaled as the figure is scaled, rounded to four decimals.

Columns in this table

Column	What it holds
Site	The short row label the figure draws.
PTM_Collapse_key	The whole identifier the values were read under, accession included. Short labels are not guaranteed to be unique; these are.
one column per sample	Headed condition_n, holding the value the cell is coloured by.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available,

for its saved settings and preprocessing recipe:

```python
conditions = ["WT", "GRK"]
fig = phossite_on_prot_regulation(data3, meta2, "Q12888",
                                  conditions=conditions, id=False)
```

```r
conditions <- c("WT", "GRK")
fig <- phossite_on_prot_regulation(data3, meta2, "Q12888",
                                   conditions = conditions, id = FALSE)
```

The name is misleading: nothing in this function is a regulation or a test. It draws exactly the heatmap on this page — one row per phosphosite of the protein, one column per sample of the chosen conditions, missing values masked — and value_type, cluster_rows, cluster_cols, filter_M2 and id are **Value Type**, **Cluster Rows**, **Cluster Columns**, **Filter *_M1** and **Toggle ID** under other names. Three of those defaults differ from the page. filter_M2 — which despite its name is the _M1 filter — starts *off* where the page has it on, so the call above draws all 45 rows of Q12888 with their multiplicity suffixes rather than the 43 in the figure. id starts on, so the sample identifiers are added unless the call turns them off. And the protein is matched as a *substring* of Protein_group, where the page's selector matches exactly, so a short accession can pull in more than one protein group.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the browser scales that render to fit the page. The default is taller than on the other single-protein pages because the number of rows is the number of sites. The table downloads separately as CSV: it holds the values behind the colours.

7 Further reading

Wu et al., absolute phosphorylation stoichiometries (2015) explains the distinction between phosphorylation stoichiometry and protein abundance. Row colours in this plot are not measurements of occupancy.

Behind this page

Function	What it produces
single_protein_heatmap_plot	The figure: scaling, clustering and colour scale

Behind this page

Function	What it produces
single_protein_heatmap_table	The matrix under it
drop_incomplete_features	The completeness filter clustering runs behind
phossite_on_prot_regulation	The same plot in the package, for a notebook